

Recovery service vault

Recovery Service Vault

backup

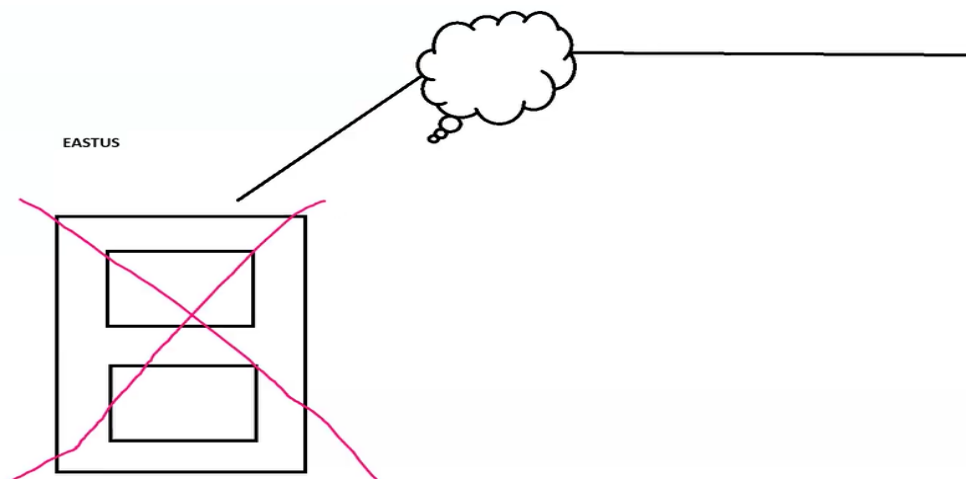
image level backup

file level backup

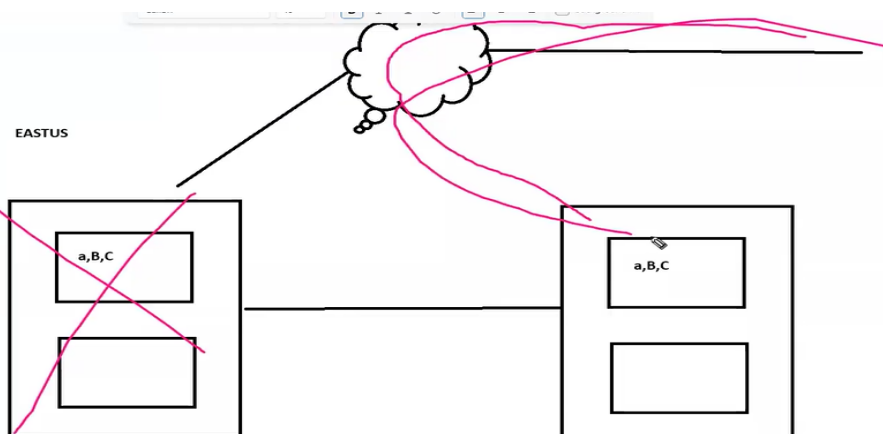
Disaster recovery

Site recovery

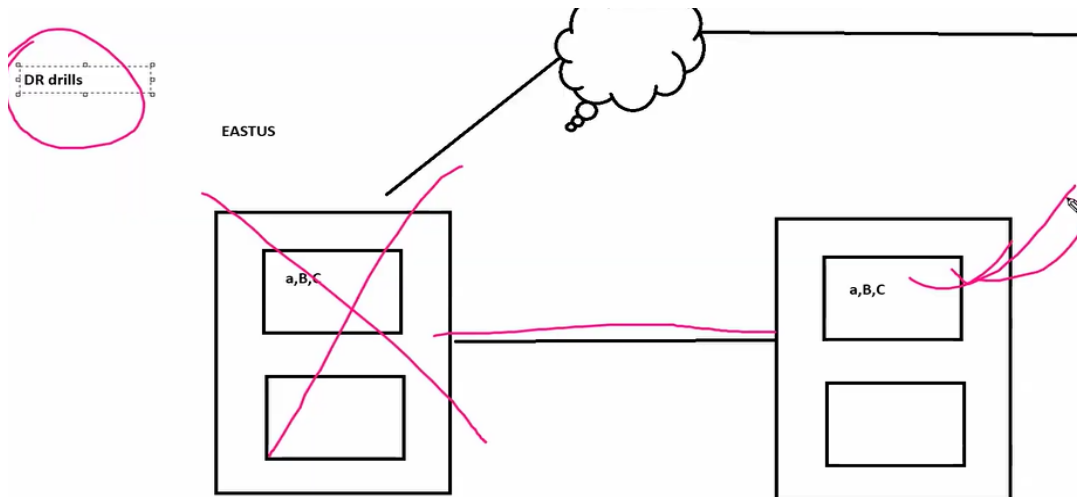
Disaster recovery works same like MySQL database (Replica)



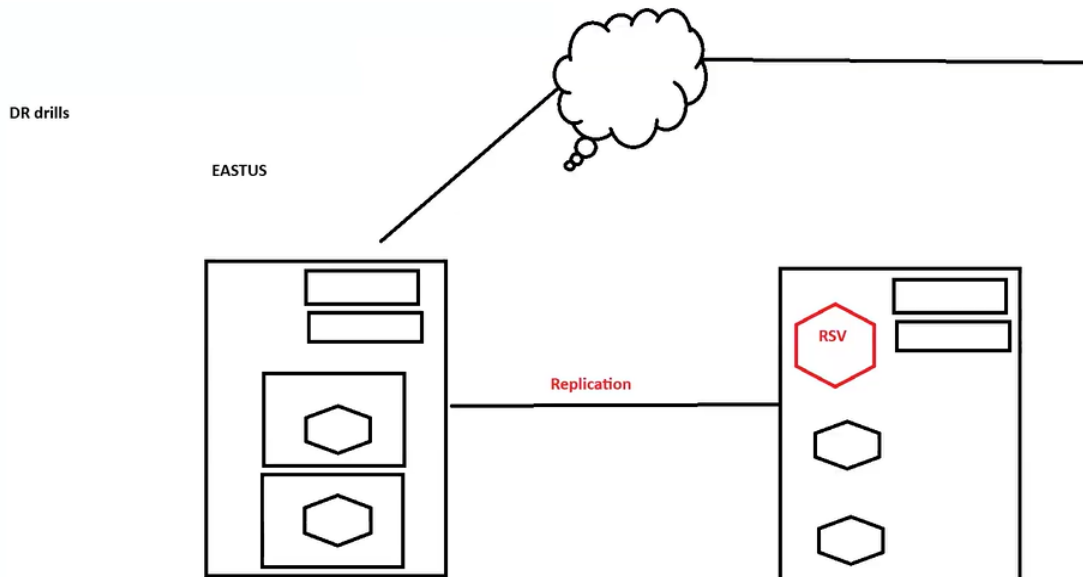
- ✓ Replicating same to another region and traffic will redirect to that when disaster happens. (**On-premise to Cloud**).



- ✓ Disaster won't happen regularly, so for testing purposes organisations conduct some **DR drills** for every **3-6 months** is data replicating or not by stopping primary servers.
- ✓ if we don't check and any disaster happens the data won't replicate to secondary then our data will be lost.

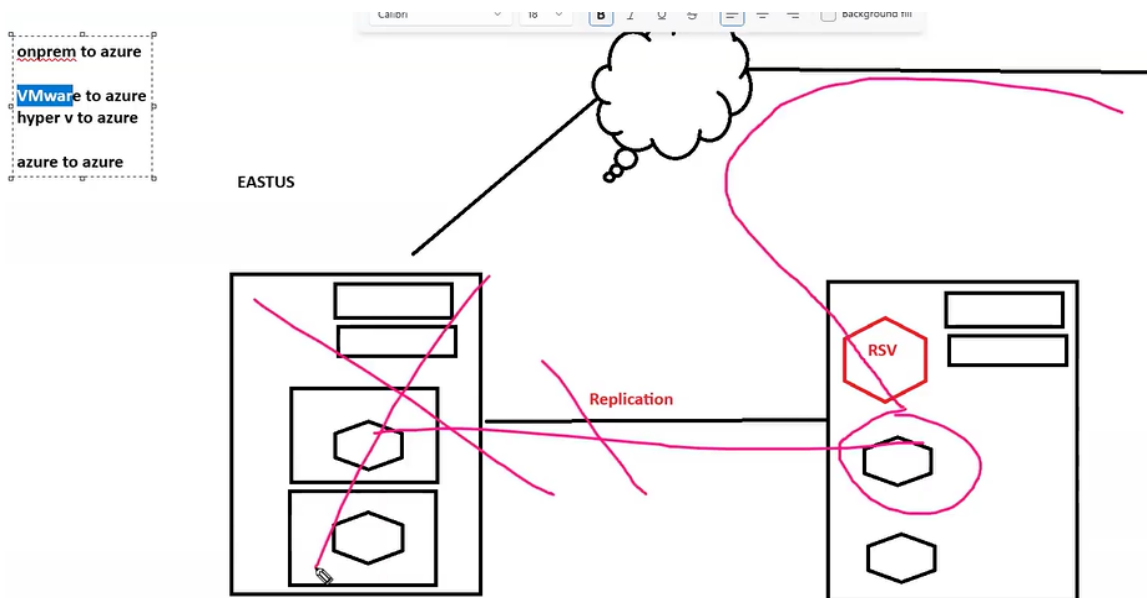


- ✓ Here in azure, we have to enable it, everything will be taking care by azure only.



- ✓ To replicate we need Rg & subnet and that should be replicate on target area
- ✓ RSV (Recovery service vault) will be created on target region, then we will enable the replication.
- ✓ After replication, Disk from primary will replicate to secondary disk only, not server level we can see above diagram.

- Azure to Azure
- Onprem to Azure
 - VMwar to Azure
 - Hyper v to Azure



- ✓ By using different tools, we can migrate onprem to azure
- ✓ Before tools they migrated using above VMwar,Hyper all these.

Note: We want **DR drills** and **cost optimization** we use this DR activity.

- If it is database we should do replica on DB to DB only not on servers

In lab purpose

- ✓ **First create Rg and VM in Central US**
- ✓ **Next create RSV in Central India**

Home > CreateVm-MicrosoftWindowsServer.WindowsServer-202-20260226172253 | Overview

UREPvm Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search Help me copy this VM in any region

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Monitor Resource visualizer Connect Networking Network settings Load balancing

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback CLI / PS

Essentials

Resource group (move): [UREPVM-RG](#)

Status: Running

Location: Central US

Subscription (move): [Azure subscription 1](#)

Subscription ID: 5d75b66d-66bf-44e0-8d7e-7e61e4b043d7

Operating system: Windows (Windows Server 2025 Datacenter)

Size: Standard_DC1ds_v3

Primary NIC public IP: [20.9.87.205](#)
[1 associated public IPs](#)

Virtual network/subnet: [vnet-centralus/snet-centralus-1](#)

DNS name: [Not configured](#)

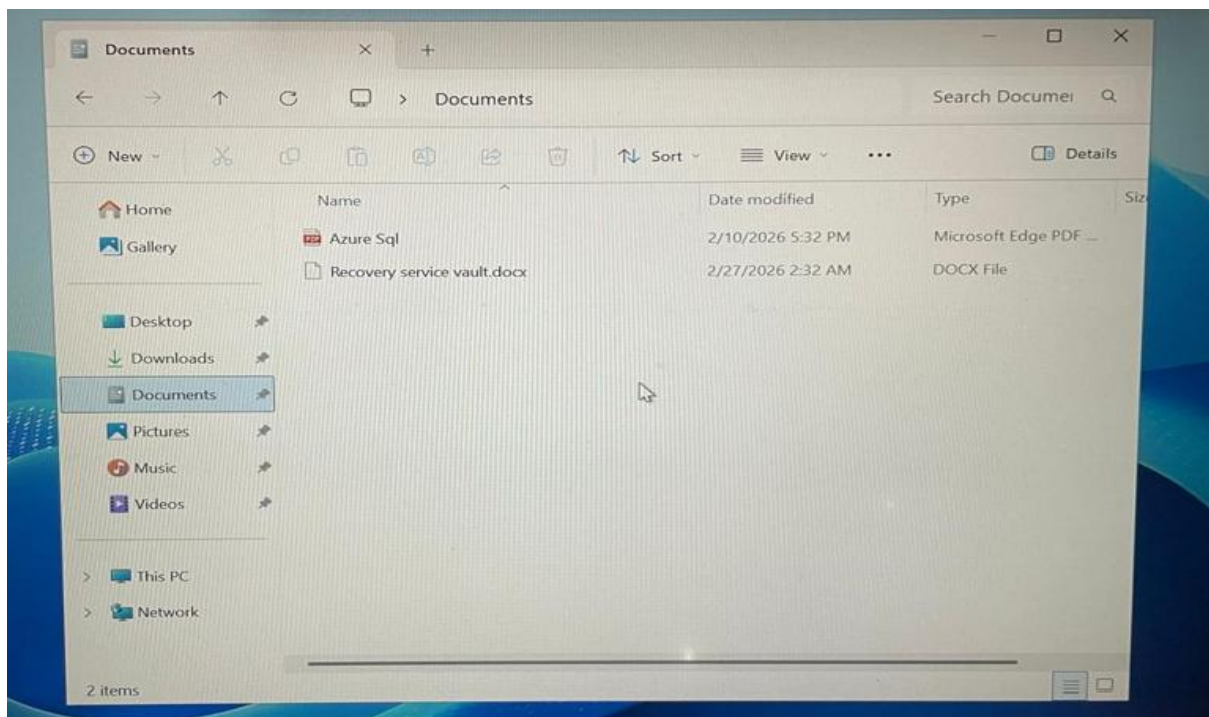
Health state: -

Time created: 26/02/2026, 11:55 UTC

Tags (edit): [Add tags](#)

[JSON View](#)

- ✓ **Before replication add any data in Central-US VM, and after doing replication and failover check whether the same data is available or not in another (Central India) VM.**



Create Recovery Services vault

- * Basics
- Redundancy
- Encryption
- Vault properties
- Networking
- Tags
- Review + create

Project Details

Select the subscription and the resource group in which you want to create the vault.

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

(New) UcentralREP-RG

Create new

Instance Details

Vault name * ⓘ

URSV216

Region * ⓘ

Central India

 Cross Subscription Restore is enabled by default for all vaults. Visit vault 'Properties' to disable the same. [Learn more.](#)

- Review + create
- Next: Redundancy
-  Feedback

Create Recovery Services vault

-  Help me create a vault
- Copy an existing vault settings

- * Basics
- Redundancy
- Encryption
- Vault properties
- Networking
- Tags
- Review + create

Backup Storage Redundancy

Locally-redundant

Locally-redundant

Zone-redundant

Geo-redundant

Create Recovery Services vault

* Basics Redundancy Encryption Vault properties Networking Tags **Review + create**

Summary

Basics

Subscription	Azure subscription 1
Resource group	UcentralREP-RG
Vault name	URSV216
Region	Central India

Redundancy

Backup Storage Redundancy	Locally-redundant
---------------------------	-------------------

Vault properties

Immutability	Disabled
--------------	----------

Networking

Connectivity method	Allow public access from all networks
---------------------	---------------------------------------

Create

Previous: Tags

Feedback

[Download a template for automation](#)



Microsoft.RecoveryServicesV2-1772108032128 | Overview

Deployment

Search

× <<



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : Microsoft.RecoveryServicesV2-1772108032128... Start time : 26/02/2026, 17:43:59
Subscription : [Azure subscription 1](#) Correlation ID : a33180f6-b90f-488a-8baa-8a2e34568812
Resource group : [UcentralREP-RG](#)

> Deployment details

> Next steps

[Go to resource](#)

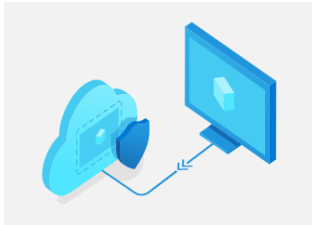
✓ **In RSV, select site recovery and choose Azure VM and click on Enable Replication below**

URSV216 | Site Recovery ☆ ...
Recovery Services vault

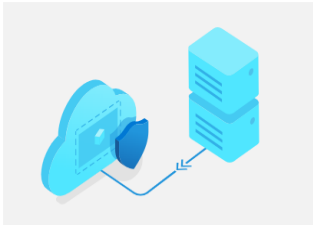
Search < < Feedback

[Learn more.](#)

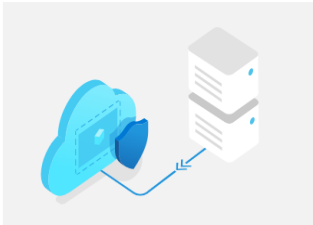
- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- > Settings
- ✓ Getting started
 - Backup
 - Site Recovery**
 - > Protected items
 - > Manage
 - > Monitoring



Azure virtual machines
Protect your Azure virtual machines for disaster recovery by replicating to another Azure region.
[1: Enable replication](#)
[2: Manage recovery plans](#)



VMware machines to Azure
Protect your VMware virtual machines for disaster recovery by replicating to Azure.
[1: Prepare infrastructure](#)
[2: Enable replication](#)
[3: Manage recovery plans](#)



Hyper-V machines to Azure
Protect your Hyper-V virtual machines for disaster recovery by replicating to Azure.
[1: Prepare infrastructure](#)
[2: Enable replication](#)
[3: Manage recovery plans](#)

✓ Give source details

Enable replication ...

- 1 Source** 2 Virtual machines 3 Replication settings 4 Manage 5 Review

Region * ⓘ

Subscription * ⓘ

Resource group * ⓘ

Virtual machine deployment model * ⓘ ☐ Classic ☒ Resource Manager

Disaster recovery between availability zones? * ⓘ ☐ Yes ☒ No

✓ Select VM

- ✓ Source **2 Virtual machines** 3 Replication settings 4 Manage 5 Review

Unable to view / select your VMs? Click [here](#) to know why.

Filter items...

Name	Virtual network	Tags
☒ UREPvm	vnet-centralus	

Showing 1 - 1 of 1 results.

Selected machines: 1

Enable replication ...

✓ Source ✓ Virtual machines **3 Replication settings** ④ Manage ⑤ Review

Location and Resource group

Target location * ⓘ

Target subscription *

Target resource group * [Create new](#)

Network

Failover virtual network * [Create new](#)

Failover subnet *

Storage

Storage is configured separately for each virtual machine. [View/edit storage configuration](#)

Availability options

Availability options are configured separately for each virtual machine. [View/edit availability options](#)

Capacity reservation

Reserve a capacity at the destination location - Central India
[Why to reserve capacity at the destination location?](#)

[Previous](#) [Next](#)

✓ **After replica if any update to do Manually, we will select manual below**

✓ Source ✓ Virtual machines ✓ Replication settings **4 Manage** ⑤ Review

Replication policy

Replication policy * ⓘ [Create new](#)

Optionally, create a replication group consisting of virtual machines running the same workload to generate multi-vm consistent recovery points.

i Enabling multi-VM consistency can impact workload performance and should only be used if machines are running the same workload and you need consistency across multiple machine

Replication group ⓘ [Create replication group](#)

Extension settings

Update settings

☐ Allow ASR to manage ⓘ
☒ Manage manually

Automation account

ⓘ
[Create](#)

✓ Click enable replication below

Enable replication ...

✓ Source ✓ Virtual machines ✓ Replication settings ✓ Manage **5 Review**

Summary

Region	Central US
Virtual machine deployment model	ResourceManagement
Subscription	Azure subscription 1
Resource group	UREPVM-RG
Availability zones	-

Virtual machine selection summary

Virtual machines to replicate 1

VM settings

Region	Central India
Target subscription	Azure subscription 1
Target resource group	UREPVM-RG-asr
Post-failover virtual network	vnet-centralus-asr
Replication policy	24-hour-retention-policy
Extension settings	Manage manually

Previous

Enable replication

✓ Below it will take 1hr to deploy we can check in site recovery jobs.

Home > URSV216

URS216 | Site Recovery jobs ☆ ...

Recovery Services vault

Search

resource visualizer

Settings

Getting started

- Backup
- Site Recovery** ☆

Protected items

Manage

Monitoring

- Alerts
- Metrics
- Diagnostic settings
- Advisor recommendations
- Backup Jobs
- Site Recovery jobs**
- Backup Alerts

Filter

Export Jobs

Feedback

Filter items...

Name	Status	Type	Item	Start Time
Create a site	InProgress	Server	asr-a2a-default-centralus	26/02/2026, 18:11:14
Create replicati...	Succeeded	ProtectionProfile	24-hour-retention-policy	26/02/2026, 18:11:09

Notifications

More events in the activity log →

Dismiss all

Deployment in progress... Running X
Deployment to resource group 'UcentralREP-RG' is in progress.
5 minutes ago

Deployment succeeded X
Deployment 'Microsoft.RecoveryServicesV2-1772108032128' to resource group 'UcentralREP-RG' was successful.
Go to resource Go to resource group
31 minutes ago

₹4,279.25 credit remaining X
Subscription 'Azure subscription 1' has a remaining credit of ₹4,279.25.
Upgrade to a Pay-As-You-Go subscription.
56 minutes ago

- ✓ Now in target region it will create Site and RG, Vnet, Disk and all the data will be replicated to that disk.
- ✓ If we don't have data, it will take 1-2hrs to replicate and with data it takes weeks for the first time.
- ✓ Below deployment is done and replica is configured to target region.

Home > URSV216

URSV216 | Site Recovery jobs

Recovery Services vault

Search resource visualizer

Settings

Getting started

Backup

Site Recovery

Protected items

Manage

Monitoring

Alerts

Metrics

Diagnostic settings

Advisor recommendations

Backup Jobs

Site Recovery jobs

Backup Alerts

Site Recovery events

Filter items...

Name	Status	Type	Item	Start Time
Enable replicati...	Succeeded	ProtectionEntity	urepvm	26/02/2026, 18:20:01
Associate replic...	Succeeded	ProtectionProfile	24-hour-retention-policy	26/02/2026, 18:17:16
Associate replic...	Succeeded	ProtectionProfile	24-hour-retention-policy	26/02/2026, 18:15:20
Map Networks	Succeeded	Network	vnet-centralus-asr	26/02/2026, 18:15:14
Map Networks	Succeeded	Network	vnet-centralus	26/02/2026, 18:14:40
Create protecti...	Succeeded	ProtectionCont...	asr-a2a-default-centralindia-cont...	26/02/2026, 18:14:40
Create a site	Succeeded	Server	asr-a2a-default-centralindia	26/02/2026, 18:13:01
Create protecti...	Succeeded	ProtectionCont...	asr-a2a-default-centralus-contai...	26/02/2026, 18:13:00
Create a site	Succeeded	Server	asr-a2a-default-centralus	26/02/2026, 18:11:14
Create replicati...	Succeeded	ProtectionProfile	24-hour-retention-policy	26/02/2026, 18:11:09

Notifications

More events in the activity log →

Dismiss all

Deployment succeeded

Deployment 'Microsoft.RecoveryServices.EnDR-202612618112' to resource group 'UcentralREP-RG' was successful.

Pin to dashboard Go to resource group

2 minutes ago

Deployment succeeded

Deployment 'Microsoft.RecoveryServicesV2-1772108032128' to resource group 'UcentralREP-RG' was successful.

Go to resource Go to resource group

45 minutes ago

₹4,279.25 credit remaining

Subscription 'Azure subscription 1' has a remaining credit of ₹4,279.25.

Upgrade to a Pay-As-You-Go subscription.

an hour ago

Home > URSV216

URSV216 | Replicated items

Recovery Services vault

Search resource visualizer

Settings

Getting started

Backup

Site Recovery

Protected items

Backup items

Replicated items

Manage

Monitoring

Alerts

Refresh Replicate Columns Filter Feedback

Last refreshed at: 26/02/2026, 18:37:02

Finished loading data from service.

Filter items...

Name	Replication Health	Status	Active location	Failover Health
UREPvm	Healthy	0% Synchronized	Central US	-

Showing 1 - 1 of 1 results.

- So, disk is created in target region
- If we want to do failover and create VM on Target disk do failover, then we can see central VM in Virtual Machines.

Click on above and we get below.

✓ Click on failover

Home > URSV216 | Replicated items



Search

[Failover](#) [Test Failover](#) [Cleanup test failover](#) [Commit](#) [Resynchronize](#) [Change recovery point](#) [Re-protect](#) [Disable Replication](#) ...

Overview

Essentials

[JSON View](#)

General

- [Properties](#)
- [Compute](#)
- [Network](#)
- [Disks](#)

Health and status

Replication Health ✓ Healthy
Status Protected
RPO 5 mins [As on 26/02/2026, 18:48:22]

Failover readiness

Last successful Test Failover ⚠ Never performed successfully
Configuration issues ✓ No issues
Agent version ✓ 9.66.7691.1
Agent status ✓ Healthy

Latest recovery points

Click above to see the latest recovery points.

Errors(0)

No errors

[Open in new page](#)

Events - Last 72 hours(0)

No events

[Open in new page](#)

Failover

UREPvm

ⓘ No test failover has been performed in past 180 days! It is recommended to perform a test failover before a failover. Skipping validation of the replication via test failover may lead to data loss or unpredictable downtime.

Benefits of doing a test failover

- Validate the replication and disaster recovery strategy without any data loss or downtime.
- A test failover doesn't impact ongoing replication, or the production environment.
- Test failover can be run on specific virtual machine (VM), or on a recovery plan containing multiple VMs.

[Learn more about benefits of doing a test failover](#) [🔗](#)

☒ I understand the risk of skipping test failover. Continue failover.

[Continue failover](#)

- ✓ **When we click failover disk will be created in target and if we enable below option SHUT DOWN Machine it will auto delete the US machine.**

Home > URSV216 | Replicated items > UREPvm

Failover ...
UREPvm

Failover direction

Source ⓘ Central US

Destination ⓘ Central India

Failover settings

Recovery point * ⓘ

Latest processed (lowest RTO) (26/02/2026, 18:48:18) (1 out of 1 disks) ▼

☐ Shut down machine before beginning the failover

Failover Cancel

Notifications ✕

[More events in the activity log →](#) [Dismiss all](#) ▼

- 

Starting the task to perform unplanned failover of virtual machine 'UREPvm'... ✕

The operation is in progress.

a minute ago
- 

Starting the task to perform unplanned failover of virtual machine 'UREPvm'... ✕

The operation is in progress.

a minute ago

Below we can see both machines but for target machine there is no public IP, let's create PubIP and associate to VM-central india.

Home

Compute infrastructure | Virtual machines

View virtual machines with critical alerts Identify VMs with network connectivity issues List VMs by security posture

Search Virtual machines Get started

Overview All resources Favorites Monitoring (Preview) Infrastructure Virtual machines Virtual Machine Scale Set (VMSS) Compute Fleet Disks + images Capacity + placement Related services Monitoring+Policy Help

Create Reservations Manage view Refresh Export to CSV Open query Assign tags Start Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...
azlinuxvm01	Azure subscript...	tejaswi-koreace...	France Central	Running	Linux	Standard_D2s_v3	51.103.92.191
azmonvm01	Azure subscript...	tejaswi-koreace...	Korea Central	Running	Windows	Standard_D2s_v3	40.82.136.208
UREPvm	Azure subscript...	UREPVM-RG-asr	Central India	Running	Windows	Standard_DC1d...	-
UREPvm	Azure subscript...	UREPVM-RG	Central US	Running	Windows	Standard_DC1d...	20.9.87.205

✓ Click below configure at Public IP address for central india

Home > Compute infrastructure | Virtual machines > UREPvm

UREPvm | Network settings

What are the requirements for attaching and detaching network interfaces? List all my network interfaces for this VM

Search List all my network interfaces for UREPvm. What are the requirements for attaching or detaching a network interface?

Attach network interface Detach network interface View topology Troubleshoot Refresh Give feedback

Network interface / IP configuration
urepvm761 (primary) / ipconfig1 (primary)

Essentials

Network interface	: urepvm761	Load balancers	: 0 (Configure)
Virtual network / subn...	: vnet-centralus-asr / snet-centralus-1	Application security g...	: 0 (Configure)
Public IP address	: - (Configure)	Network security group	: -
Private IP address	: 172.18.0.4	Accelerated networking	: Enabled
Admin security rules	: 0 (Configure)	Effective security rules	: 0

✓ Click on Add below

Home > Compute infrastructure | Virtual machines > UREPvm | Network settings > urepvm761

urepvm761 | IP configurations

Network interface

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

IP configurations

DNS servers

Network security group

Properties

Locks

Monitoring

IP Settings

Enable IP forwarding

Virtual network

Subnet *

snet-centralus-1 (172.18.0.0/24) 250 free IP addresses

250 free IP addresses

Private and public IP addresses can be assigned to a virtual machine's network interface controller. You can add as many private and public IPv4 addresses as necessary to a network interface, within the limits listed in the Azure limits article. Private IP address prefix types can only be created as a secondary IP configuration. [Learn more](#)

+ Add Make primary Delete

Name	IP Version	Type	Private IP Address	Public IP Address
☐ ipconfig1	IPv4	Primary	172.18.0.4 (Dynamic)	-

✓ Enable Public IP address settings below and add

Home > Compute infrastructure | Virtual machines > UREPvm | Network settings > urepvm761

urepvm761 | IP configurations

Network interface

Search

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IP Settings

Enable IP forwarding

Virtual network

Subnet *

snet-centralus-1 (172.18.0.0/24) 250 free IP address

Private and public IP addresses can be assigned to a virtual machine's network interface controller. You can add as many private and public IPv4 addresses as necessary to a network interface, within the limits listed in the Azure limits article. Private IP address prefix types can only be created as a secondary IP configuration. [Learn more](#)

+ Add Make primary Delete

Name	IP Version	Type	Private IP Address	Public IP Address
☐ ipconfig1	IPv4	Primary	172.18.0.4 (Dynamic)	-

Add IP configuration

*** Updating network interface

Updating network interface 'urepvm761'.

Private IP address settings

Private IP address type

IP address

IP address prefix (Only for IPv4)

Allocation

Dynamic

Static

Public IP address settings

Associate public IP address

Public IP address *

(New) pip-vnet-centralus-asr-centralindi... Create a public IP address

Add Cancel

Give feedback

✓ Pub-ip is created, lets associate with the VM-centralindia

+ Add Make primary Delete

Name	IP Version	Type	Private IP Address	Public IP Address
☐ ipconfig1	IPv4	Primary	172.18.0.4 (Dynamic)	-
☐ UcentralindiaPip	IPv4	Secondary	172.18.0.5 (Dynamic)	104.211.72.175 (pip-vnet-centralus-asr-centralindia-snet-centralus-1)

✓ Search Public IP Address and you see below

✓ Click on it and open and first De-Associate and again Associate.

Search

Virtual network

Virtual Network overview

Virtual networks

NAT gateways

Public IP addresses

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

	Name ↑	IP address	IP version	SKU	Associated to
☒	pip-vnet-centralus-asr-centralindi	20.204.3.233	IPv4	Standard	-
☐	UREPvm-ip	20.9.87.205	IPv4	Standard	urepvm761

✓ De-Associate and Associate

Home > Network foundation | Public IP addresses

pip-vnet-centralus-asr-centralindia-snet-centralus-1 Public IP address

Can you check if my public Ip is highly available How to

Search

Associate Dissociate Delete Move Refresh Open in mobile Give feedback

Overview

Activity log

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Tags

Resource visualizer

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Automation

Essentials

Resource group (move) : UREPVM-RG-asr

Location (move) : Central India

Subscription (move) : Azure subscription 1

Subscription ID : 5d75b66d-66bf-44e0-8d7e-7e61e4b043d7

See more

Tags (edit) : Add tags

SKU : Standard

Tier : Regional

IP address : 20.204.3.233

DNS name : -

Associated to : -

Virtual machine : -

✓ Give below details when you click on Associate.

Home > Network foundation | Public IP addresses

pip-vnet-centralus-asr-centralindia-snet-centralus-1 Public IP address

Can you ch

Search

Associate Dissociate Delete Move Refresh Open in mob

Overview

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Help

Essentials

Resource group (move) : UREPVM-RG-asr

Location (move) : Central India

Subscription (move) : Azure subscription 1

Subscription ID : 5d75b66d-66bf-44e0-8d7e-7e61e4b043d7

See more

Tags (edit) : Add tags

Get Started Properties Tutorials

Associate public IP address

pip-vnet-centralus-asr-centralindia-snet-centralus-1

Choose the resource to which you want to assoc

Resource type

Network interface

Network interface * ⓘ

urepvm761

resource group: UREPVM-RG-asr

✓ Know when we refresh Pub-IP is associated to Central India VM

Microsoft Compute infrastructure | Virtual machines

Search

Virtual machines Get started

Overview All resources Favorites Monitoring (Preview) Infrastructure Virtual machines Virtual Machine Scale Set (VMSS) Compute Fleet

+ Create Reservations Manage view Refresh Export to CSV Open query Assign tags Start Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...
BR-vm	Azure subscript...	BR-rg	Central India	Running	Linux	Standard_DC1d...	4.213.163.226
UREPvm	Azure subscript...	UREPVM-RG-A...	Central India	Running	Windows	Standard_DC1d...	20.204.3.233
UREPvm	Azure subscript...	UREPVM-RG	Central US	Stopped (deall...	Windows	Standard_DC1d...	20.9.87.205
VM1	Azure subscript...	VRG1	Australia East	Running	Windows	Standard_DC1d...	20.58.189.41

✓ Copy Pub-IP of Central India and login, Check if the data is available are not.

✓ Before login give NSG rules to connect.

✓ Create NSG and associate to VM-central India.

Home > UcentindiaNSG

UcentindiaNSG | Network interfaces

Search

Associate Refresh Dissociate

Search network interfaces

Name ↑↓	Public IP address ↑↓	Private IP address ↑↓	Virtual machine ↑↓
urepvm761	20.204.3.233	172.18.0.4	UREPvm

Home > Compute infrastructure | Virtual machines > UREPvm

UREPvm | Network settings

Search

Activity log Access control (IAM) Tags Diagnose and solve problems Resource visualizer Settings Inbound security rules Outbound security rules Network interfaces

Network interface : urepvm761

Virtual network / subn... : vnet-centralus-asr / snet-centralus-1

Public IP address : 20.204.3.233

Private IP address : 172.18.0.4

Admin security rules : 0 (Configure)

Load balancers : 0 (Configure)

Application security g... : 0 (Configure)

Network security group : UcentindiaNSG

Accelerated networking : Enabled

Effective security rules : 0

Rules Collapse all

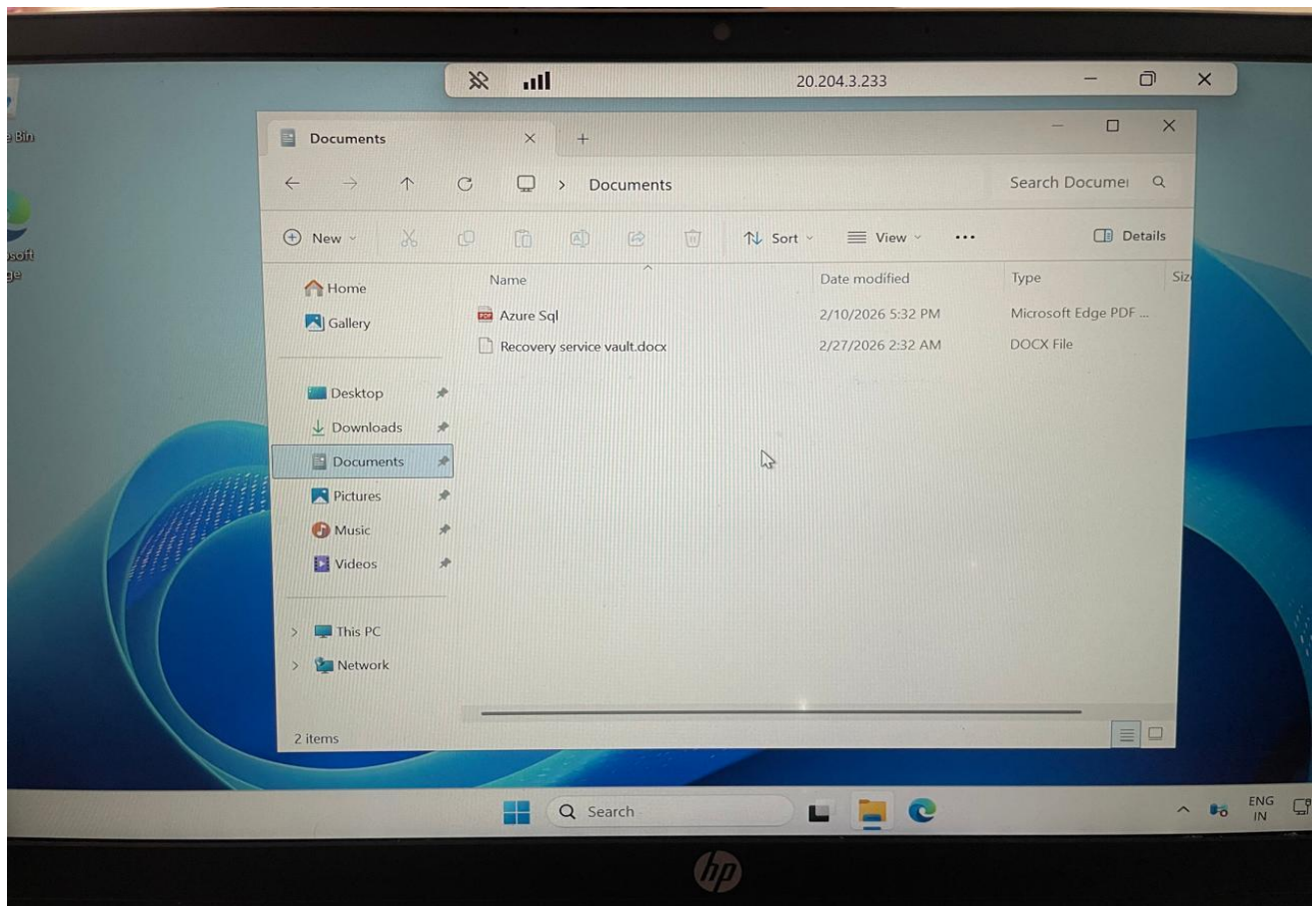
Network security group UcentindiaNSG (attached to networkInterface: urepvm761)

Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
100	allow-RDP	3389	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow

- ✓ **Login to central india VM and check the data is available.**



- ✓ **We backed up from primary to secondary successfully**
- ✓ **Whatever the data stored in Central-US VM, after disaster we restored from US to India by doing failover.**